

The Distinguished Determinant Zero Mode: Exact Coarsening Invariance, Signed-Prefix Duality, and a Cancellation Obstruction

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Abstract

For the literal post-bin determinant coefficient $A_X(n)$, the distinguished frequency is

$$Z_X = \sum_n A_X(n).$$

We first prove that Z_X is exactly invariant under every provenance-preserving partition and coarsening of the native atoms: determinant binning can change energy but cannot create zero-mode cancellation. We then insert the literal fixed- h_0 affine Möbius split. On each ordered physical fiber the remaining sign is

$$\sigma(r) = \mu^2(L_1(r))\mu^2(L_2(r))\lambda(L_1(r))\lambda(L_2(r)).$$

If $S(k) = \sum_{i \leq k} \sigma(r_i)$, discrete summation by parts gives the sharp dual identity

$$\sup_{\|W\|_{\text{BV}^*} \leq 1} \left| \sum_i \sigma(r_i) W(r_i) \right| = \max_k |S(k)|.$$

Thus the exact structure-free norm controlling the weighted zero mode is a signed-prefix discrepancy, not determinant energy, support density, or nonzero-frequency dispersion.

For the actual outer synthesis we obtain a fully explicit sufficient criterion involving its literal coefficients, fiber variations, signed-prefix discrepancies, and content remainder. The criterion is fiberwise sharp for uniform control over the declared weight class; after summing outer keys it is sufficient and need not be necessary. No power saving is proved for the growing physical signs. This is an L0/L1 reduction and obstruction theorem. It does not establish the exponent $\eta_Z > 0$, a parity breakthrough, or a prime-pair conclusion.

Keywords: distinguished zero mode; determinant frequency; signed-prefix discrepancy; summation by parts; fixed shift.

1 The distinguished frequency

Fix a prescribed $h_0 \neq 0$. Let $\mathcal{T}_X$ be the canonical native archive and $\nu : \mathcal{T}_X \rightarrow \mathcal{N}_X$ the determinant map. For literal coefficients u_t , define

$$A_X(n) = \sum_{\nu(t)=n} u_t, \quad Z_X = \sum_{n \in \mathcal{N}_X} A_X(n), \quad D_X = \sum_{n \in \mathcal{N}_X} |A_X(n)|^2.$$

This is the normalized determinant-DFT frequency $r = 0$ of TPC-32, not an orbit or transfer-operator zero mode [1, 2].

Theorem 1.1 (Exact coarsening invariance). *Let $\pi : \mathcal{T}_X \rightarrow Y$ be any finite partition map, and form*

$$C_\pi(y) = \sum_{\pi(t)=y} u_t.$$

Then

$$\boxed{\sum_{y \in Y} C_\pi(y) = \sum_{t \in \mathcal{T}_X} u_t = Z_X.}$$

In particular, determinant binning, content binning, or any further exact coarsening preserves the distinguished zero mode.

Proof. The fibers of π form a disjoint partition of $\mathcal{T}_X$, so the finite sums may be interchanged. $\square$

Remark 1.2 (Energy behaves differently). The identity in [theorem 1.1](#) has no analogue for $\sum_y |C_\pi(y)|^2$: coarsening creates cross terms and can either raise or lower energy. Hence an energy census and a zero-mode estimate are logically independent.

2 Literal affine sign decomposition

TPC-90 resolves a primitive physical branch into affine forms

$$L_{1,\theta,X}(r) = d_{\theta,X} + s_{\theta,X}r, \quad L_{2,\theta,X}(r) = u_{\theta,X} + a_{\theta,X}r$$

with the same prescribed determinant

$$s_{\theta,X}u_{\theta,X} - a_{\theta,X}d_{\theta,X} = h_0.$$

Assume, on the declared block, that content, masks, both polarizations, and all native outer keys have been retained and that the literal outer reassembly has been verified. Its zero-mode interface then has the form

$$Z_X = \sum_{\theta \in \Theta_X} c_{\theta,X} \sum_{r \in \mathcal{I}_{\theta,X}} W_{\theta,X}(r) \sigma_{\theta,X}(r) + E_{\text{content},X}, \quad (1)$$

where

$$\sigma_{\theta,X}(r) = \mu^2(L_1(r))\mu^2(L_2(r))\lambda(L_1(r))\lambda(L_2(r)) \in \{-1, 0, 1\}.$$

This is the exact split $\mu = \mu^2\lambda$ [4]. Local squarefree density determines where σ vanishes, but not the signs on its active support.

3 Sharp signed-prefix duality

We now prove the energy-free analytic statement used by the literal zero-mode route.

Definition 3.1 (Terminal variation norm). For an ordered finite list $w = (w_1, \dots, w_N)$, define

$$\|w\|_{\text{BV}_*} = |w_N| + \sum_{k=1}^{N-1} |w_k - w_{k+1}|.$$

Theorem 3.2 (Exact Abel duality). *Let $\sigma_1, \dots, \sigma_N \in \mathbb{C}$, and put*

$$S_k = \sum_{i=1}^k \sigma_i.$$

Then every $w \in \mathbb{C}^N$ satisfies the exact identity

$$\sum_{i=1}^N \sigma_i w_i = S_N w_N + \sum_{k=1}^{N-1} S_k (w_k - w_{k+1}),$$

and

$$\sup_{\|w\|_{\text{BV}_*} \leq 1} \left| \sum_{i=1}^N \sigma_i w_i \right| = \max_{1 \leq k \leq N} |S_k|.$$

Proof. Put

$$d_k = w_k - w_{k+1} \quad (k < N), \quad d_N = w_N.$$

Then $w_i = \sum_{k=i}^N d_k$, so

$$\sum_i \sigma_i w_i = \sum_k S_k d_k.$$

Also $\|w\|_{\text{BV}_*} = \sum_k |d_k|$. The upper bound is ℓ^1 - ℓ^∞ duality. Equality is obtained by placing all d -mass at an index where $|S_k|$ is maximal, with the conjugate phase of S_k . $\square$

Corollary 3.3 (Literal fiber bound). *Order one actual interval $\mathcal{I}_{\theta,X} = \{r_1 < \dots < r_N\}$, and define*

$$\Delta_{\theta,X} = \max_{1 \leq k \leq N} \left| \sum_{i=1}^k \sigma_{\theta,X}(r_i) \right|, \quad V_{\theta,X} = \|(W_{\theta,X}(r_i))_{i=1}^N\|_{\text{BV}_*}.$$

Then

$$\left| \sum_{r \in \mathcal{I}_{\theta,X}} W_{\theta,X}(r) \sigma_{\theta,X}(r) \right| \leq \Delta_{\theta,X} V_{\theta,X},$$

with optimal constant for the declared variation norm.

Remark 3.4 (The ordering is part of the statement). The order is the literal translated integer order on the physical fiber. Reordering signs to improve their prefixes changes the coefficient and is not permitted.

4 A complete energy-independent go criterion

Theorem 4.1 (Outer zero-mode certificate). *Assume the exact literal reassembly (1). Then*

$$|Z_X| \leq \sum_{\theta \in \Theta_X} |c_{\theta,X}| \Delta_{\theta,X} V_{\theta,X} + |E_{\text{content},X}|.$$

Consequently, the desired estimate

$$|Z_X| \leq X^{-\eta_Z + o(1)} Q_X^2$$

follows if the right-hand side has that size.

Proof. Apply [corollary 3.3](#) to every term of (1), then use the triangle inequality. $\square$

Remark 4.2 (Sufficient, not necessary for one physical weight). The criterion allows no cancellation between different outer keys, so it may be stronger than the scalar target. For uniform estimates over all weights with prescribed variation, however, [theorem 3.2](#) shows that the prefix discrepancy is the exact operator norm. A future proof may instead exploit literal outer cancellation, but it must retain the same coefficients and provenance.

Proposition 4.3 (Unsigned data cannot pass the criterion). *If the Liouville product in $\sigma_{\theta,X}$ is replaced by the constant sign 1, then*

$$\Delta_{\theta,X} = \#\{r_i : \mu^2(L_1(r_i))\mu^2(L_2(r_i)) = 1\}.$$

Thus squarefree-support density, by itself, gives the full active mass rather than cancellation.

Proof. Every nonzero summand is +1, so the prefix sums are nonnegative and maximal at the terminal prefix. $\square$

5 Why determinant energy does not estimate the zero mode

For a vector $A = (A(n))_{n \in \mathcal{N}}$, Cauchy–Schwarz gives only

$$|Z(A)|^2 \leq |\mathcal{N}| D(A).$$

It is sharp for a constant vector. On the same support and with the same moduli, roots of unity can have zero sum. This finite nonidentifiability was isolated in TPC-82 [3]. TPC-97 further shows that constant-preserving projection preserves Z while reducing energy, so it cannot manufacture a better numerator [5].

Theorem 5.1 (Independent-gate verdict). *None of the following data implies a positive exponent η_Z for the actual growing coefficient:*

- (i) *determinant support and multiplicities;*
- (ii) *the value or a lower bound of D_X ;*
- (iii) *unsigned squarefree density;*
- (iv) *cancellation at every nonzero determinant-DFT frequency; or*
- (v) *exact provenance-preserving coarsening.*

A proof of $\eta_Z > 0$ must estimate the literal signed prefixes, exploit a comparably strong outer cancellation, or introduce another arithmetic theorem that controls the same scalar Z_X .

Proof. Items (i)–(ii) are defeated by aligned and root-of-unity vectors with the same support and moduli. Item (iii) is [proposition 4.3](#). A constant signal has no nonzero Fourier frequencies but has maximal zero mode, proving (iv). Item (v) is [theorem 1.1](#). $\square$

6 Audited conclusion and next gate

Statement	Status
Coarsening invariance and Abel duality	Exact finite theorems (L0).
Literal affine-sign crosswalk	Exact provided the TPC-90 outer reassembly is verified on the declared block (L1).
Bound for the actual $\Delta_{\theta,X}$	Not proved; parity-facing arithmetic input.
Positive exponent η_Z	Not established.
Determinant energy or physical endpoint	Independent and not charged here.

TPC-111 therefore produces a sharp *criterion*, not the desired cancellation estimate. TPC-112 may combine a future η_Z with a future determinant exponent λ_D , but it must record the current values as “unproved” and keep the physical endpoint ledger separate.

Explicit exclusions. No theorem here controls a growing two-form Liouville or Möbius correlation at fixed h_0 . Nothing is specialized to $h_0 = 2$. The paper proves no L2 saving, parity breach, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem.

References

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